

A Pulsed, Broadband NMR Spectrometer*

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A pulsed nuclear magnetic resonance spectrometer suitable for measurements in solids is described. The spectrometer is phase coherent, and the pulse timing is synchronous with the operating frequency of the spectrometer. The apparatus uses a versatile pulse programmer and a gated broadband rf power amplifier which delivers 1.6 kW at 60 MHz with a 3-db bandwidth of 5 to 90 MHz. With the exception of the probe, the spectrometer does not contain any tuned elements. This permits short (0.5 μ sec) recovery times between the end of the rf pulse and the start of the signal. The spectrometer relies heavily upon the use of commercially available modules. A detailed description of the spectrometer construction and a discussion of typical operating characteristics are presented.

I. INTRODUCTION

This article discusses the construction and operation of a pulsed nuclear magnetic resonance (NMR) spectrometer suitable for the study of pulsed NMR measurements in solids. The design objective of the spectrometer described in this article was the building of a simple yet versatile apparatus with a bandwidth suitable for many types of pulsed NMR experiments in solids. Some of the major features achieved in this spectrometer are (a) the ability to operate at any frequency between 5 and 90 MHz without any tuned circuits, except for the sample coil, (b) versatile pulse programming capability, (c) synchronization of the pulse programmer and spectrometer operating frequency, (d) rf pulse output power of 1.6 kW, and (e) receiver recovery times as short as 0.5 μ sec. The spectrometer also makes extensive use of commercially available devices in order to simplify construction and improve the reliability of the apparatus.

The block diagram of the spectrometer is shown in Fig. 1, and a detailed description of the apparatus is provided in Section II through VIII. The block diagram has been labeled by the section number of the article in which that portion of the spectrometer is discussed. The description of the apparatus is followed by a discussion of the typical operation and performance of the spectrometer in Section IX.

II. PULSE PROGRAMMER

A. Timing Simulator

A versatile pulse programmer greatly enhances the range of experiments which may be executed by a pulsed NMR apparatus. Among its other necessary attributes for work

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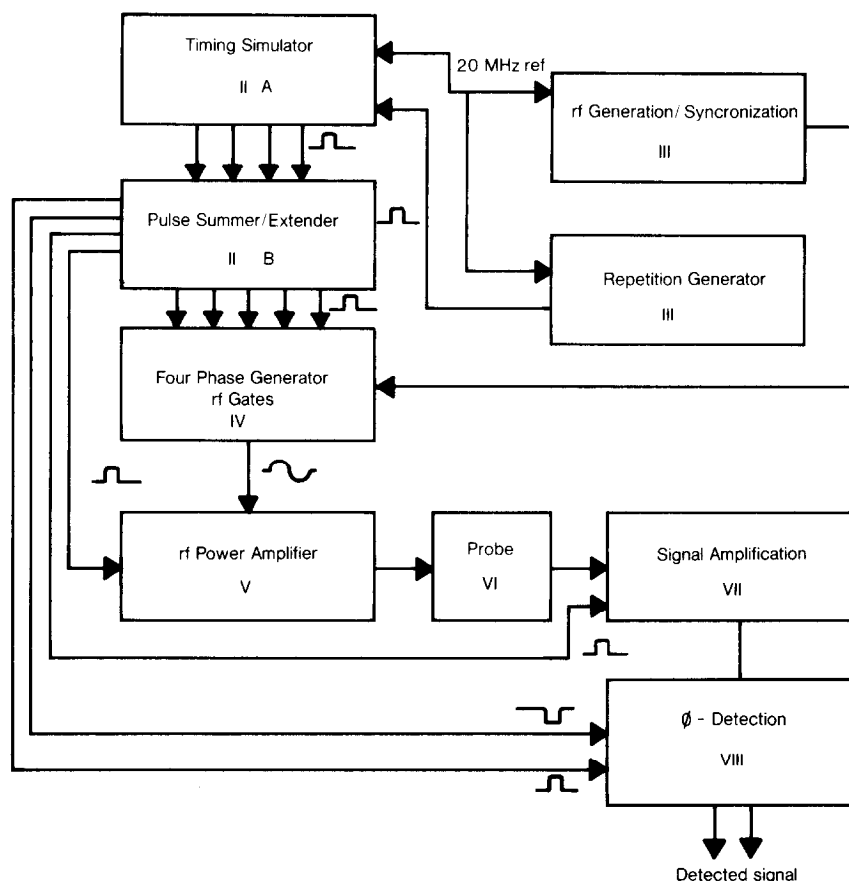


FIG. 1. Block diagram of spectrometer. Individual blocks are labeled to correspond to the section of the paper in which they are discussed.

in solids, the pulse programmer should be capable of generating a large variety of pulse sequences, ranging from single pulse to the complicated N-pulse sequences used in certain line-narrowing experiments (1). The pulse programmer should also be flexible enough to permit rapid incorporation of new pulse sequences. Unfortunately, "homemade" pulse programmers usually require a sacrifice of versatility in order to simplify design and/or programming methods. Several such designs have appeared in the literature (2, 3). A second alternative is to use computers to control pulse programming (4). Although the variety of pulse sequences which can be generated in such an arrangement is almost unlimited, it is an expensive alternative. Computer-controlled pulse programming also requires someone to undertake the nontrivial task of interfacing the computer to the spectrometer.

A third approach, which is a cross between the above two methods, uses a commercially available device called a Timing Simulator (TS) (Interface Technology, Inc., 852 North Cummings Rd., Covina, Calif. 91724) as a pulse programmer. The TS provides the spectrometer with sufficient versatility for most commonly used pulse sequences and is easily programmed. A pulse sequence to be executed by the TS is

stored in consecutive addresses of a bipolar random access memory (RAM). Each address contains level information (pulse on or off) for each of the eight independent output channels, as well as timing information instructing the TS as to the length of time it is to reside at a particular memory address. For our system, this time may range from 50 nsec to 999 msec, in increments as small as 50 nsec. Upon actual execution of a pulse sequence, the TS sequentially accesses each memory address, holding the output level information for the programmed length of time. Upon execution of the last memory address (lma) of the program, the TS returns to the start of the program and either repeats the program immediately or waits for an external trigger pulse to repeat the program, depending on the position of a front-panel toggle switch.

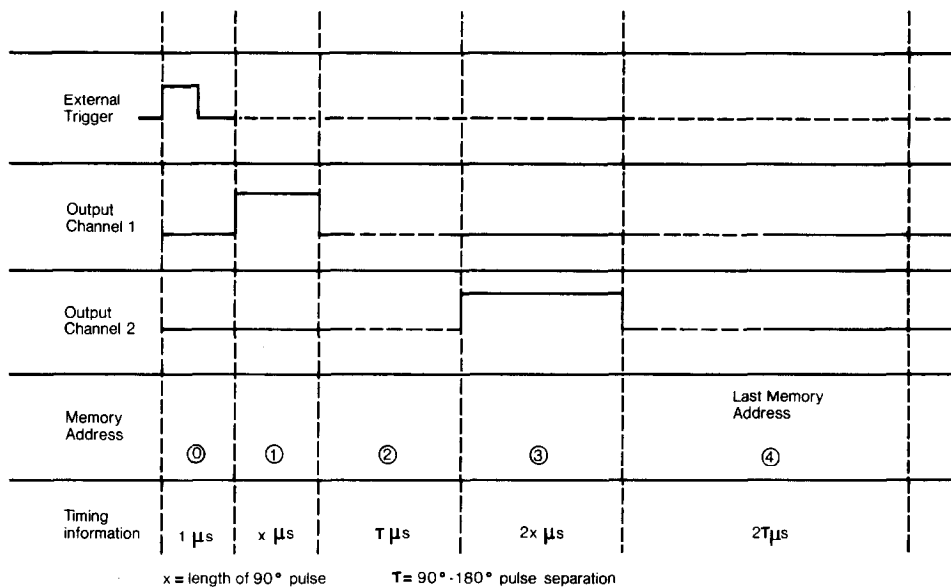
The TS derives its time base from an internal 20-MHz crystal oscillator, thus all timing intervals are crystal controlled. The TS may also be operated from an external, TTL compatible 20-MHz reference frequency, as is done in the spectrometer described here. The gating pulses can then be made coherent with the rf. The reasons for synchronizing the gating pulses and rf are discussed in Section III.

The eight output channels are all TTL compatible and have an output impedance of $50\ \Omega$. This allows them to drive the other sections of this spectrometer, without the connecting coaxial cables degrading the quality of the pulses.

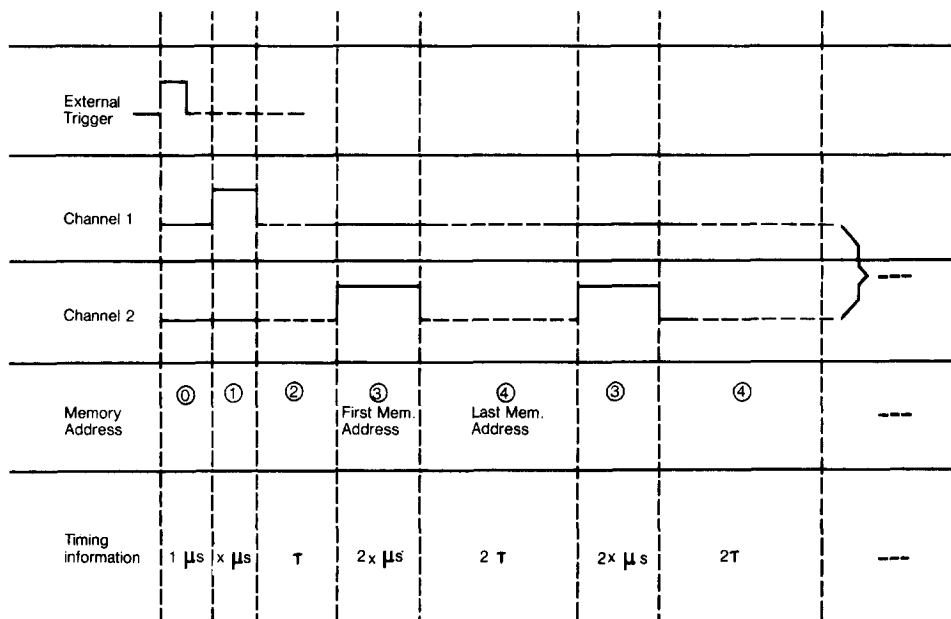
A typical pulse sequence program for a simple 90° , τ , 180° pulse sequence is illustrated in Fig. 2A. Programming consists of drawing the desired pulse sequence and assigning memory addresses as done in Fig. 2A. In this case, memory address four is the lma used in this program, and this address is loaded into the TS as the last memory address to be executed in this pulse program. One then proceeds to load the timing and level information into the TS using front-panel toggle and thumb-wheel switches. Although the program in Fig. 2A uses only two channels, the remaining six channels may be programmed to generate pulses during any of the five memory addresses used in the program. Programmed level information for a given memory address is present for the programmed time in each output channel, with a maximum skew or nonsynchronism between channels of 5 nsec.

An extremely useful feature of the TS, referred to as dynamic updating, permits one to change level and/or timing information at any memory address while the pulse program is being executed. In reference to the pulse sequence of Fig. 2A, one may vary the 90° to 180° pulse separation, τ , simply by setting new timing information for memory address two, then depressing the dynamic update switch. The new timing information is loaded into the memory during the pulse sequence in progress when the switch is depressed and is then executed for all subsequent pulse sequences. By using the dynamic update feature, one can modify a pulse sequence while an experiment is in progress without having to reprogram each memory address used in the pulse program or stopping the experiment.

The loop cycle counter (lcc) is a very useful option available with the basic TS that permits the execution of long pulse sequences in which a portion of the sequence is repetitive in nature. The lcc permits the TS to repeat a portion or all of a given pulse sequence between 1 and 1000 times. The use of the lcc option is illustrated in Fig. 2B, in which the pulse sequence of Fig. 2A is used to generate a 90° , τ , $N(180^\circ, 2\tau)$ pulse sequence such as is used in a Carr, Purcell, Meiboom, and Gill (5) determination of T_2 . The lcc is used by programming memory address three as the "first memory address"



A



B

FIG. 2. (A) A simple 90° , τ , 180° , τ , echo pulse sequence; (B) A CPMG pulse sequence demonstrating the use of the single-cycle loop counter.

(fma), and by programming the lcc to execute N loops, where N is between 1 and 1000 and is the number of 180° pulses desired. Upon execution of the sequence, the TS will first step through memory addresses zero through four, then repeat addresses three (fma) and four (lma) ($N - 1$) more times before returning to memory address zero to await another external trigger pulse. For certain experiments, it may be desirable to repeat an entire pulse sequence rather than just a portion of it. This may be accomplished by designating memory address zero as the fma or by setting a switch on the TS which automatically tells the TS to return to the fma rather than address zero after completing N loops.

The TS used in the spectrometer discussed here has an expanded memory option (256 programmable memory addresses). This expanded memory can be used to store programs of several frequently used pulse sequences. By selecting the appropriate first and last memory addresses, the pulse sequence required for a particular experiment can be executed, while pulse sequence programs at memory addresses other than those between the fma and lma are stored for future use.

The TS as used in this spectrometer is operated in conjunction with an external repetition rate generator which provides a pulse to trigger the TS to perform the programmed pulse sequence once and then wait for the next trigger pulse. The details of the repetition rate generator are described in Section III. The TS may be programmed to generate its own repetition rate by setting the duration of the lma and setting the appropriate front-panel switch, but this mode of operation limits the variety of pulse sequences which can be performed by the TS, especially when the lcc option is used.

B. Pulse Summer and Extender

Frequently, a pulse programmer must generate, for a given pulse sequence, a number of coincident (or practically coincident) pulses to be used by various components of a spectrometer other than the rf gates. Such devices, in this apparatus, include the gated rf power amplifier (Section V) and the receiver gates (Section VII). Furthermore, it is often desirable to delay or advance the timing of the pulses controlling these devices relative to the pulses used to gate the rf, so as to account for the differing rise and fall times in these various devices. The TS, alone, could be used for this purpose, but this increases the complexity of the program required for any given pulse sequence. The circuit described in this section, called a pulse summer and extender (PSE), is used to generate, from a single TS pulse, a set of delayed or advanced pulses to be used by the devices referred to above. Although the PSE is not an essential part of the spectrometer, it greatly simplifies the setup and modification of a pulse sequence executed by the TS.

A second purpose of the PSE is to extend the length of pulses from the TS in order to circumvent the TS's inherent 50-nsec pulse width resolution. This feature permits the precise pulse length setability required when short, high-power rf pulses are used in an experiment.

A schematic diagram of the PSE is shown in Fig. 3. The circuit accepts pulses from the TS, incident at inputs 1 through 4. The rising edge of the pulse at point A (or, equivalently, the trailing edge of the input pulse) is used to trigger a monostable multivibrator (MM1), whose period can be set with an external timing resistor and capacitor. These timing components are shown in the inset of Fig. 3. The pulse generated by MM1 is summed with the original pulse at point B to extend the length of the pulse

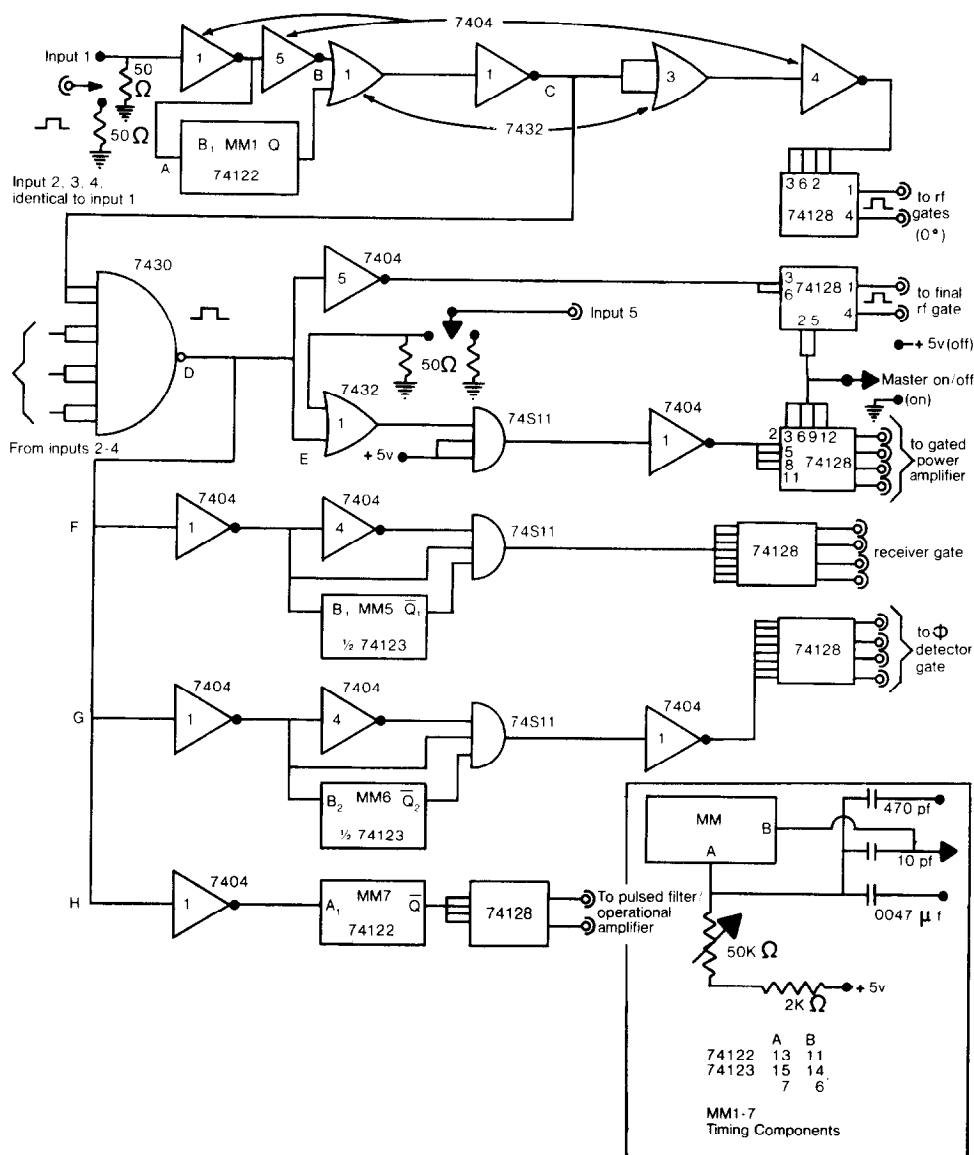


FIG. 3. Pulse length summer and extender circuit. The inverter and OR function logic symbols show the number of functions (not chips) connected in series to obtain the appropriate delay.

from the TS. The original pulse at point B has been delayed (by ~ 50 nsec) by the inverter string at the input, so that the rise of the extension pulse from MM1 occurs slightly before the fall of the delayed original pulse. Following point B, the extender pulse is further delayed by the OR gate and inverter strings. The final output pulse is obtained from a 50-Ω line driver (74128), and is used to operate an rf gate (Section IV). The four separate pulse inputs and outputs of the PSE are used for turning on and off, four separate rf gates with independent input rf amplitudes and phases. This will be described in a later section.

Pulses at points C of input channels 1 through 4 are summed at point D and are used to generate gating pulses for the devices listed below:

- (1) final rf gate (Section IV);
- (2) gated rf power amplifier (Section V);
- (3) receiver gate (Section VII);
- (4) phase-sensitive detector gate (Section VIII);

and

- (5) pulsed filter/amplifier preconditioning (Section VIII).

Since the rf gating pulses are delayed between point C and their output points, the pulses generated for the devices listed above occur somewhat earlier (by ~ 35 nsec), so the small propagation delays and rise times of these various devices can be compensated for.

Pulses for the final rf gate and gated rf power amplifier are of the same duration as the pulses used for the rf gates. The pulse for the final rf gate is delayed by the inverter string between point D and its output, so that it is coincident with pulses for the initial rf gates. Pulses for the gated power amplifier may be summed with pulses from an auxiliary input (input 5) for the purpose of preconditioning the power amplifier, as explained in Section V. It is not necessary to delay the pulse to the power amplifier because of the delays already inherent in the gate system of the amplifier.

Pulses for the receiver gate and the phase sensitive detector gate are generated at points F and G, respectively. The purpose of each of these sections is twofold. The first is to generate pulses which begin slightly before the rf gating pulse. The second is to extend these pulses beyond the rf gating pulses (via MM5 and MM6) in order to protect the receiver and phase-sensitive detector during the "ringdown" of the probe circuit.

The pulses generated by the portion of the circuit following point H are used to control devices which process the phase-sensitive detected signal, as discussed in Section VIII.

The PSE, as shown in Fig. 3, uses strings of TTL gates (inverters or OR gates) to generate the appropriate delays between different pulse outputs. These delay strings may be replaced by digital delay devices (Data Delay Devices, 253 Crooks Ave., Clifton, N.J. 07011), with which the amount of delay may be more easily selected. This has been done on a newer version of the PSE which is used in another spectrometer.

III. RF GENERATION AND COHERENT SPECTROMETER OPERATION

The spectrometer is designed for operation primarily at 60 MHz, although operation at any frequency between 5 and 90 MHz is possible with only slight modification (see Sections VI and VII). Since generation of a suitable rf frequency is relatively straightforward and must be tailored to suit individual requirements, only a brief discussion of the two methods used in this spectrometer will be presented. This is followed by a discussion of the need for and method(s) of rf-pulse programmer (TS) synchronization.

Ordinarily, the spectrometer obtains a continuous 60-MHz rf signal from a crystal oscillator, the output of which is also used to generate a 20-MHz reference for phase-locking the TS. When it is necessary to use a variable frequency source, the spectrometer utilizes a Hewlett-Packard (HP) (1820 Embarcadero Rd., Palo Alto,

Calif. 94303), 608F signal generator operated in conjunction with a HP8708A synchronizer. The synchronizer improves the stability of the HP608F (to ~ 0.2 ppm/10 min) and also provides a simple method of synchronizing the 608F and the TS. The 608F signal generator and 8708A frequency synchronizer are no longer manufactured by HP. Both these devices are available on the surplus market, however. A second and probably more expedient variable frequency source is a commercial frequency synthesizer, which can also be locked to the 20-MHz reference frequency.

As has been discussed in the literature (6), there is some advantage to synchronizing the pulse programmer to the rf, so that the waveform of the rf during the pulse(s) start and end at the same point on the rf waveform each time the pulse sequence is repeated.

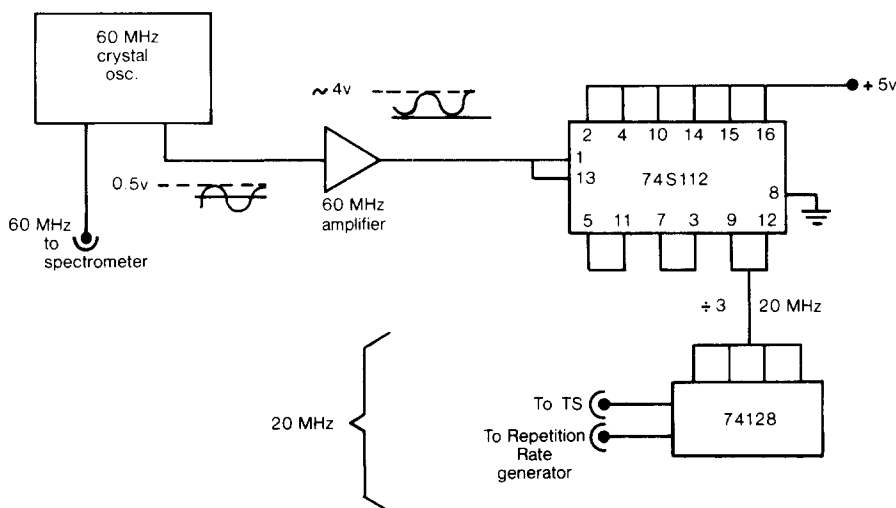


FIG. 4. Circuit for the generation of the 20-MHz reference from the 60-MHz crystal oscillator.

Failure to synchronize the pulse programmer to the rf can result in variations in the direction (phase) and magnitude of the rotation of the nuclear magnetization during an rf pulse each time the pulse(s) is repeated. This is especially true when short (high-power) rf pulses are used.

In this spectrometer, the rf and TS are operated synchronously by phase locking the 20 MHz which operates the TS to the rf frequency of the spectrometer. When the apparatus is operated at 60 MHz using the crystal oscillator, a 20-MHz signal is obtained directly from the 60 MHz as shown in Fig. 4. When the spectrometer is to be operated at a frequency other than 60 MHz, and the HP608F signal generator is used in conjunction with the HP8708A synchronizer, the rf and TS are easily synchronized because the HP8708A synchronizer phase locks the HP608F signal generator to a 20-MHz reference signal. Thus by using the 20-MHz reference controlling the TS to also operate the HP8708A synchronizer, the rf and TS are synchronized at spectrometer operating frequencies other than 20 MHz. It should be pointed out that when the HP8708A and HP608F are operated in this manner, the frequency of the HP608F may not be varied continuously, but only in steps of 100 KHz.

As was pointed out above, the synchronization of the rf and TS is required to guarantee that the effect of an rf pulse on the spin system does not change each time the pulse sequence is repeated. In order for this to occur, the repetition rate of the pulse sequence must also be synchronous with the rf and TS. In this spectrometer, the start of the pulse sequence from the TS is determined by a trigger pulse from a repetition rate

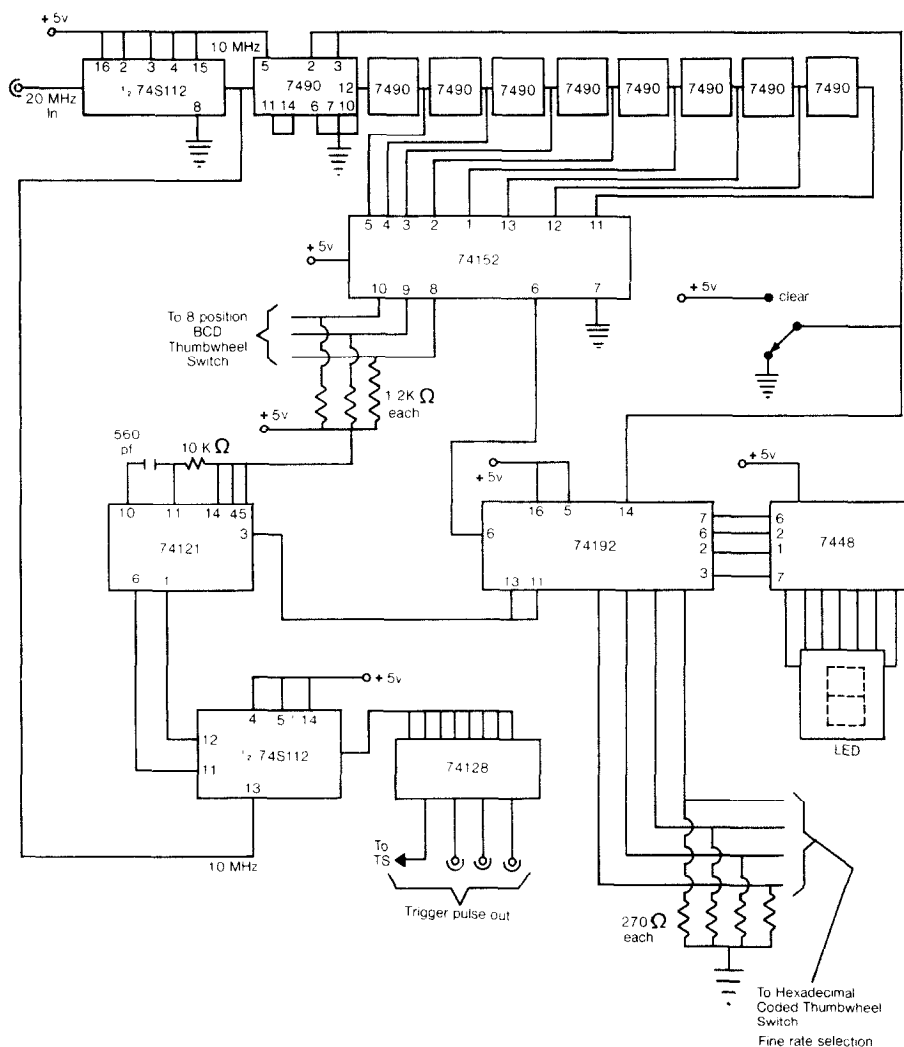


FIG. 5. Repetition rate generator used to trigger the timing simulator.

generator shown in Fig. 5. The repetition rate generator is operated synchronously with the rf and TS by using the same 20-MHz signal which phase locks the rf and TS as its time base.

This circuit uses a 74S112 dual J-K flip-flop to divide the incoming 20-MHz signal by two. The resulting 10-MHz signal is then passed to a series of nine 7490 decade dividers providing frequencies of 1 MHz to 0.01 Hz in decade steps. The output of the last eight

7490's is fed to a 74152 multiplexer, so that the output frequency desired may be selected by a specific BCD input to the multiplexer. The output of the multiplexer goes to a 74192 programmable counter which may be set to divide the input frequency by any integer between 1 and 16. The carry output is used to trigger a 74121 monostable multivibrator in order to set the triggering pulse width at about 2 μ sec. The complementary outputs of the 74121 monostable multivibrator are passed through the second half of the 74S112 dual J-K flip-flop which is clocked by the 10-MHz output of its first half. This operation synchronizes both leading and trailing edges of the triggering pulse output with the timing simulator. This synchronized pulse drives a 74128 quad 50- Ω line driver which provides the final trigger pulse output. This method of synchronizing the leading and trailing edges of a pulse by passing it through a flip-flop which is clocked coherently with the rf may be used on any pulse even if the device generating this pulse is not phase-locked to the rf.

It should be noted that, with the synchronization of the rf, TS, and repetition rate generator, the interval between frequencies which permit coherent operation of the spectrometer is determined by the repetition rate of the experiment. More specifically, the pulse sequence repetition rate in hertz is also the minimum interval between frequencies at which the spectrometer operated coherently. Thus, if the spectrometer operating frequency must be varied in the course of an experiment by units of, say 100 Hz, the maximum repetition rate which permits coherent spectrometer operation at each frequency is also 100 Hz.

The careful selection of the experiment repetition rate effects the coherence of the rf and TS in that it guarantees that the effect of the pulse sequence on the spin system is the same each time the experiment is repeated. A similar consideration must also be applied to the minimum separation of pulses within the same sequence. For example, consider a pulse sequence consisting of two rf pulses of the same length and phase, separated by a time, τ . If the spectrometer operating frequency is not an integral multiple of 20 MHz, the pulse separation required to keep the effect of each pulse on the spin system the same is determined by the interval in time separating those times when the spectrometer operating frequency and the 20-MHz TS time base *phases* coincide. For example, if the operating frequency is chosen to be 56.4 MHz, then pulse separation must be incremented in multiples of 2.5 μ sec for the two pulses to be truly identical. Thus, in pulse sequences involving the application of two (or more) identical pulses, careful selection of the pulse length separation is required to guarantee that each pulse has the same effect on the spin system.

IV. FOUR-PHASE GENERATOR AND RF GATES

For many pulsed NMR experiments, it is necessary to generate pulse sequences in which two or more of the rf pulses are obtained from continuous rf signals of the same frequency but of different relative phases. The ability to generate rf pulses differing in phase by 0, 90, 180, and 270° provides sufficient flexibility for most pulsed NMR experiments. In this spectrometer, the generation of these four rf phases is accomplished through the use of commercially available power splitters whose outputs are phase-shifted relative to one another. Other groups (7) have also used similar versions of these devices for this purpose.

The network of power splitters and double-balanced mixers shown in Fig. 6 is

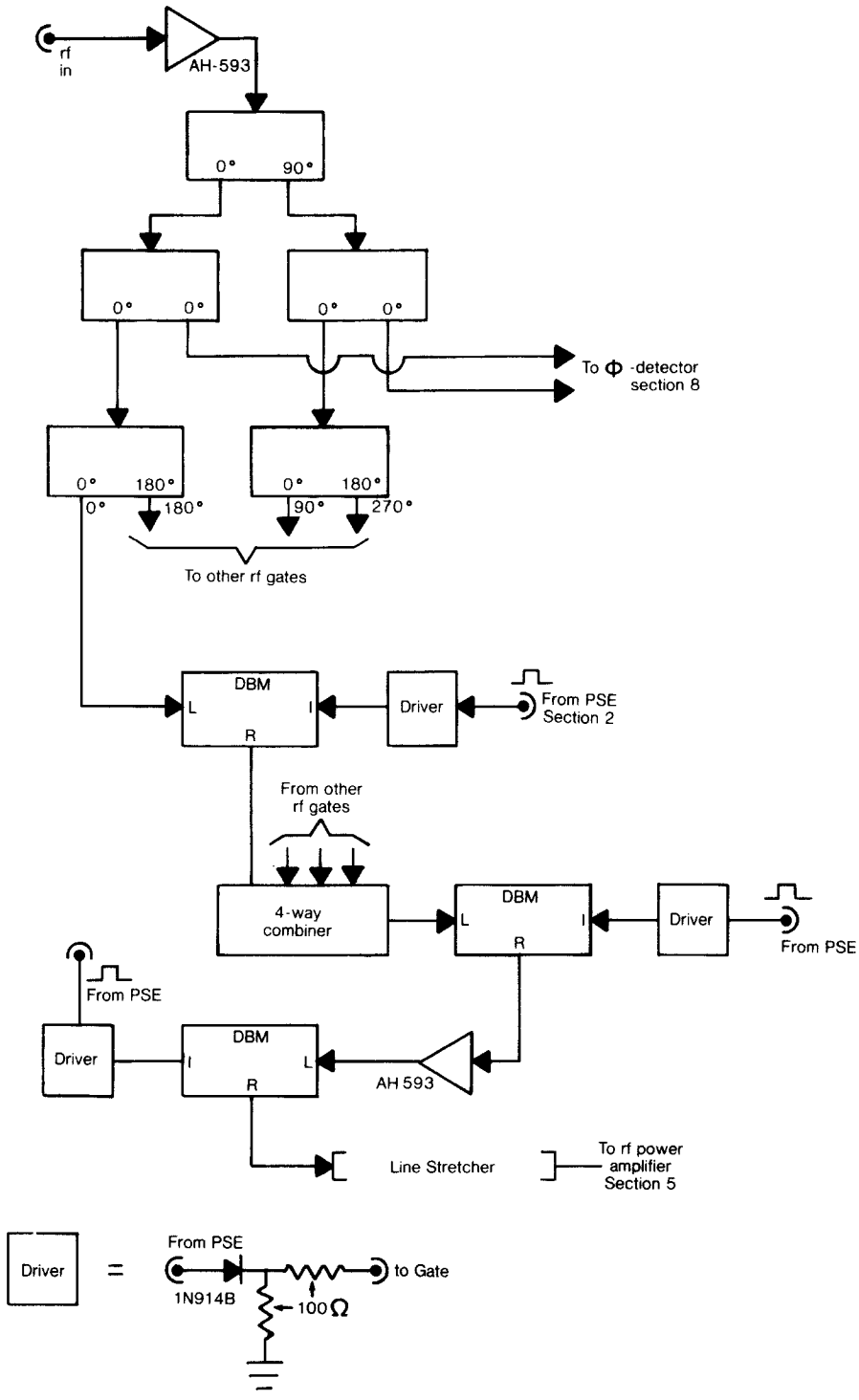


FIG. 6. Four-phase generator and rf gate assembly.

responsible for providing rf pulses of relative phases 0, 90, 180, and 270°. Each of these devices was obtained from Mini-Circuits, Inc. (837–843 Utica Ave., Brooklyn, N.Y. 11203) and is packaged individually in rf-tight containers. Interconnections between the various devices are made using matched lengths of RG58/U coaxial cable, in order to preserve the phase relationships of different rf channels as determined by the power/phase splitters. Because the four-phase generator and rf gating assembly do not involve any particularly new concepts, only the more salient features of Fig. 6 are discussed.

A continuous rf signal at the input of the circuit in Fig. 6 is first amplified by an AH-593 (Optimax, Inc., P.O. Box 105, Advance Ln., Colmar, Pa. 18915) broadband (10–400 MHz) amplifier in order to boost the signal level input to the four-phase generator. The amplifier also serves to isolate the frequency source from the four rf phases to be generated. The only component of the four-phase generator of limited bandwidth (1 octave) is the 0° to 90° power splitter following the initial amplifier. The 0° to 90° power splitter may be easily changed to another of the proper frequency range for spectrometer operation.

The four rf channels of differing phase are gated by double-balanced mixer (DBM) gates driven by the PSE circuit of Section II B, as shown in the inset of Fig. 6. The voltage divider network sets the current into the gate at ~20 mA. The diode serves to isolate the gate from the PSE between pulses. The carrier rejection of these DBM gates is not sufficient for pulsed NMR applications, so the rf pulses from each of the phase channels are summed in a four-way combiner and then passed through a second rf gate driven by a summed pulse sequence from the PSE in order to obtain a carrier rejection of >100 db. Since the four-way combiner attenuates the rf pulses by ~6 db, the rf pulses are amplified to an ~1-V peak by a final AH-593 amplifier. The output of the amplifier is then passed through a final rf gate to further improve carrier rejection and attenuate output noise of the AH-593 amplifier between pulses. The final pulse sequence is passed through a 50- Ω line stretcher (General Radio, 300 Baker Ave., Concord, Mass, 01742), so that the overall phase of the pulse sequence may be varied with respect to the phase of the rf reference signal(s) used for phase-sensitive detection (Section VIII).

V. RADIO FREQUENCY POWER AMPLIFIER

The radio frequency power amplifier for amplifying the low-level rf pulses from the gating network must be capable of delivering high-power rf pulses with very short rise and fall times. Unfortunately, these two requirements are, to some degree, mutually exclusive. In recent years, several pulse amplifier designs have been described in the literature (7, 8–10). The power amplifier in this spectrometer is capable of generating high-power (1.6 kW at 60 MHz) rf pulses with rise and fall times that are typically on the order of 50 nsec. It does not use any tuned elements, and the 3-db bandwidth of the amplifier is 5 to 90 MHz.

The first stage of this unit is a commercial, solid-state, broadband power amplifier (Model 3100L, Electronic Navigation Industries, 3000 Winton Road South, Rochester, N.Y. 14632) which can generate 100-W rf pulses from the low-level rf pulses at the output of the rf gates. These 100-W rf pulses are then used to drive a homemade, gated power amplifier.

The 3100L has input and output impedances of 50 Ω , so that it matches well into

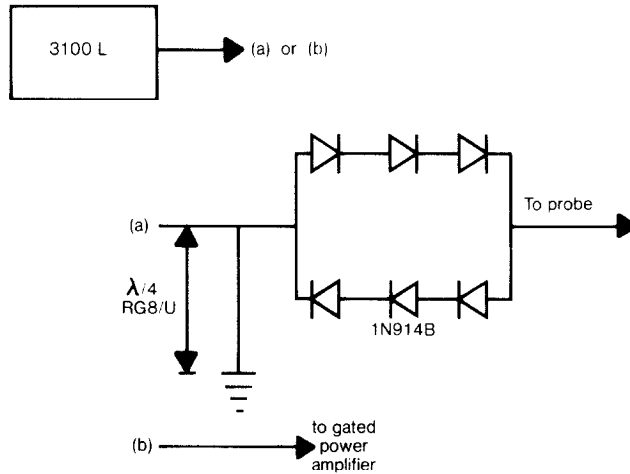


FIG. 7. (a) $\lambda/4$ short and diode noiseblock used with 3100L amplifier when rf power amplifier is not used; and (b) with use of rf power amplifier.

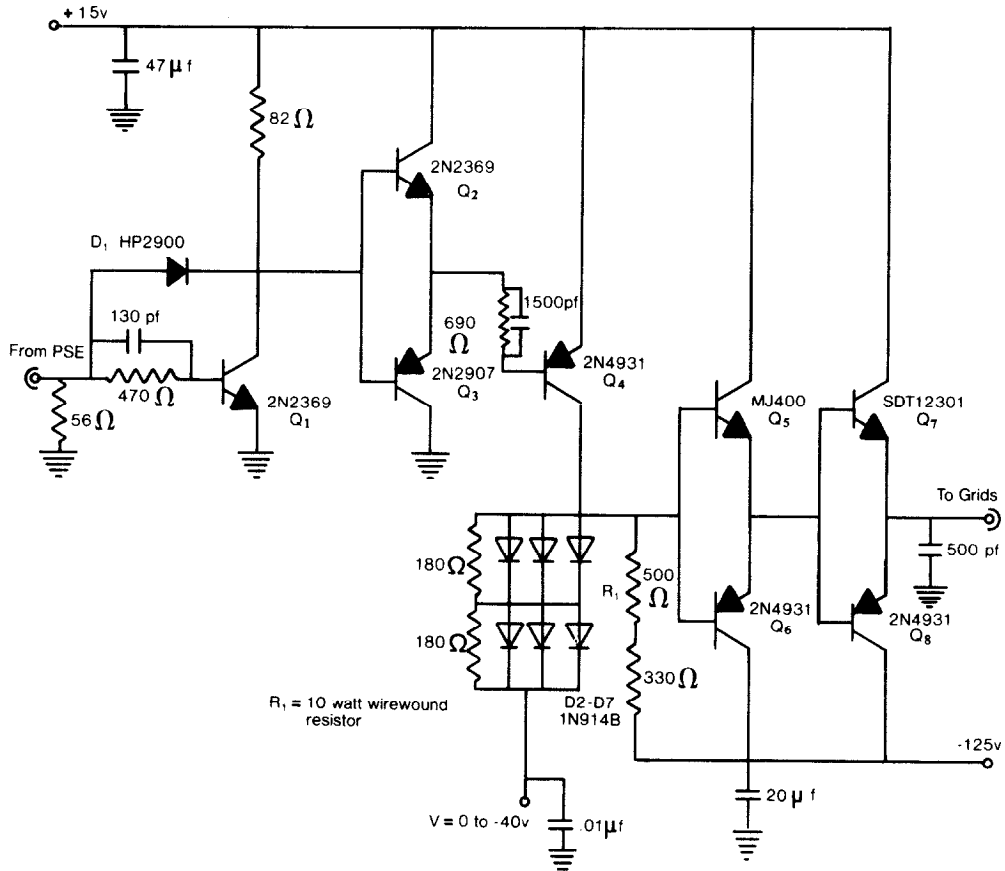


FIG. 8. Pulse amplifier circuit used to gate the 4CX600B grids of the rf power amplifier tubes.

other components of the system. The 3100L usually represents the initial stage of power amplification, but its 100-W power is often sufficient for many experiments. When the 3100L is used without the gated power amplifier discussed below, the noise output of the 3100L between pulses necessitates the use of a diode noiseblock as shown in Fig. 7. Care must be taken not to overdrive the 3100L during an rf pulse, since this results in a low-frequency ringing at its output following the rf pulse. This ringing, which can impair the recovery time of the receiver, can be greatly attenuated by placing a $\lambda/4$ short on the

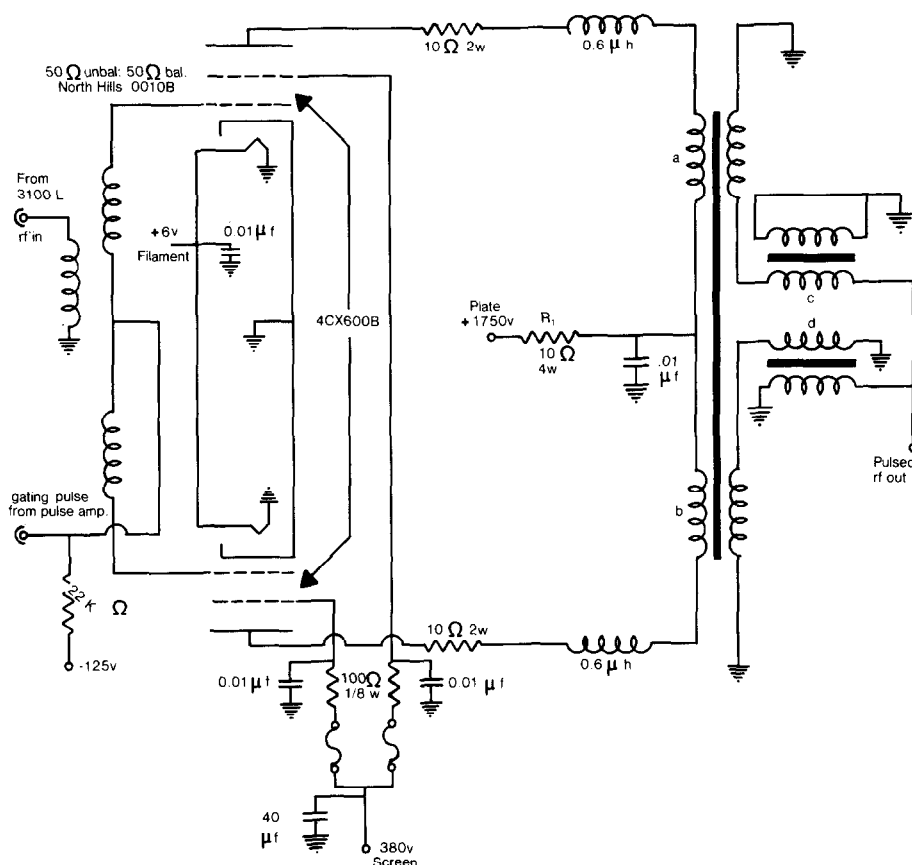


FIG. 9. The rf power amplifier.

output of the 3100L, as illustrated in Fig. 7. The $\lambda/4$ short has quite a broad bandwidth (~ 15 MHz) and does not appear to degrade the rise and fall times of the rf pulses.

The gated power amplifier that follows the 3100L is shown in Figs. 8 and 9. The amplifier uses two 4CX600B tetrodes in a push-pull configuration. The grids of these tubes are normally biased beyond cutoff and are gated on during an rf pulse by the pulse amplifier circuit of Fig. 8. The circuits of Figs. 8 and 9 are discussed individually below.

The pulse amplifier (Fig. 8) gates the transmitter by changing the grid bias on the two tubes in the transmitter from -125 V in the quiescent state, to ~ 0 V during an rf pulse.

The -125-V grid bias is sufficient to cut the tubes off between rf pulses. The rise and fall times of the grid pulses determine the rise and fall times of the final rf pulse output of the gated power amplifier, and are, typically, $\leq 50\text{ nsec}$.

The pulse amplifier accepts a standard TTL output from the PSE to drive the inverting amplifier, Q_1 , which is cut off in the quiescent state. When the input of Q_1 is switched to $\sim 2.5\text{ V}$ by the PSE, Q_1 turns on, and drives the complimentary pair, Q_2 and Q_3 . The RC network in the base of Q_1 improves the switching time of Q_1 , as does D_1 , which prevents Q_1 from reaching saturation.

The complimentary pair Q_2 and Q_3 are required to drive the RC network in the base circuit of inverting amplifier Q_4 . Q_4 is not a particularly fast transistor, and the RC network that drives it speeds up its rise and fall times. The speed of Q_4 is also enhanced by diodes $D_2\text{--}D_7$, which prevent Q_4 from saturating during the pulse. The inductance of the wirewound resistor, R_1 , was also found to improve the switching speed of Q_4 .

The output of Q_4 drives the complimentary pair Q_5 and Q_6 , which provide a low-impedance drive for the bases of the output stage, complimentary pair Q_7 and Q_8 . In the quiescent state, Q_7 is off, and the tube grids (Fig. 3) are biased beyond cutoff at -125 V . During a pulse, Q_7 conducts hard, biasing the tube grids at $\sim 0\text{ V}$. The small capacitor between Q_7 and Q_8 serves to keep the rf signal on the tube grids from leaking to the pulse amplifier. This capacitor does not appear to degrade the pulse amplifier switching time.

The gated power amplifier shown in Fig. 9 represents the final stage of rf amplification and is driven directly (no diode noise block) by the 3100L. Versions of the gated power amplifier design discussed here have been used in our laboratory for the past six years. The gated power amplifier uses two 4CX600B tetrodes in a push-pull configuration as the only active components in the design. These tubes were selected for their high transconductance and low interelectrode capacitance. An additional advantage of using tetrodes is the tube output power depends primarily on the screen voltage and is relatively insensitive to plate voltage fluctuations, such as those which may occur during a long rf pulse.

As is mentioned above, the power amplifier is operated in the gated mode by switching the tube grid bias from -125 V between rf pulses to $\sim 0\text{ V}$ during the rf pulse. The off state of the transmitter between rf pulses provides additional isolation from carrier leakage (and the 3100L noise output) of $\sim 40\text{ db}$. The tube grids are gated by the output of the pulse amplifier (Fig. 8) via the center tap in the secondary of input transformer, T1. In the quiescent state, the tube grids are biased at -125 V , and the amplifier is cut off. At the time of an rf pulse, 100-W ($50\ \Omega$) rf pulses from the 3100L are incident upon the primary of T1. At the same time, the pulse amplifier switches the tube grid bias to $\sim 0\text{ V}$, and the tubes conduct hard. The input transformer, T1, is a $50\text{-}\Omega$ unbalanced: $50\text{-}\Omega$ balanced broadband transformer (North Hills Electronics, Inc., Alexander Pl., Glen Cove, N.Y. 11542). This impedance transformation was selected in order to efficiently couple rf power to the tube grids, which have a low dynamic impedance ($\sim 12\ \Omega$) during an rf pulse.

The normal electrode operating voltages of the tubes are also shown in Fig. 9. Since the gain of the amplifier is very dependent upon the screen operating voltage, it is necessary to use a well-regulated power supply to operate the screens. Extreme care must be taken to insure that the plate voltage is maintained well above the screen

voltage whenever the tubes are on. Failure to maintain this condition causes the tube screens to draw much more than their rated maximum current (15 mA), and tube failure may result. Since the 4CX-600B used in this transmitter design is rather expensive, some precautions are taken to protect the tube screens in the event of plate voltage failure. This screen protection is provided by interlocking the screen power supply to the plate voltage supply, so that a plate voltage drop below 1000 V will remove the screen operating voltage from the tube. A less elaborate protection mechanism is to fuse the tube screens with $100\text{-}\Omega$ $\frac{1}{8}$ -W carbon resistors, as shown in Fig. 9. These resistors also serve to suppress parasitic oscillations.

The plate circuit of each tube contains a $10\text{-}\Omega$ 2-W resistor to aid in suppressing parasitic oscillations and a small inductor ($\sim 0.6\text{ }\mu\text{H}$), which in conjunction with the plate capacitance of the tube (6 pF) peaks the impedance of the plate circuit at ~ 60 MHz. This slight tuning of the plate circuit improves the gain of the power transmitter at higher frequencies.

The output coupling transformer is a $200\text{-}\Omega$ balanced : $50\text{-}\Omega$ unbalanced transmission-line transformer (11). The transmission line used is a twisted pair with an impedance of $100\text{ }\Omega$ made from 20 AWG Teflon-insulated wire. The individual sections, *a*, *b*, *c*, and *d* of the transformer (Fig. 9) are made by winding five turns of the transmission line around a toroidal ferrite core. (Ferroxcube No. 528T500-3C8, Ferroxcube Corp., 5083 Kings Hwy, Saugerties, N.Y. 12477).

Sections *a* and *b* are made from two separate sections of transmission line wrapped around opposite sides of a single ferrite core. One end of each of the primary winding of *a* and *b* is connected to a tube plate, and the opposite ends are connected together to form the center tap of the transformer. In the secondary of *a* and *b*, the ends opposite the center tap of the primary are grounded, while the opposite ends drive transformers *c* and *d*. Transformers *c* and *d* are identical to *a* and *b* except that each is wound on a separate ferrite toroid. Since, in push-pull operation, the rf signals on each of the two tubes are 180° out of phase, the phase of the rf from one of the tubes must be inverted in order to combine the power from each tube at the output. Transformer *d* inverts the phase of the rf from the tube which drives it, and transformer *c* is included to approximately match the propagation delay of transformer *d*, so that the rf signal from each tube is in phase when combined at the output.

The plate voltage is fed to the tubes through transformer sections *a* and *b* and resistor, R_1 . The resistor, R_1 , was included in order to eliminate spurious low-frequency ringing at the output of the power amplifier following an rf pulse. We find that unless this low-frequency ringing is damped, the minimum recovery time of the receiver is limited to $\sim 2\text{ }\mu\text{sec}$ following the end of the rf pulse. With the incorporation of R_1 , the power amplifier output is "clean" in less than $0.1\text{ }\mu\text{sec}$ following the rf pulse. The receiver recovery time need not be this fast, since other factors influencing its recovery time must be considered.

One problem which has been encountered with the rf power amplifier design using the 4CX-600B tubes has been the poor long pulse performance of the tubes: Tube life is drastically reduced when rf pulses longer than $\sim 500\text{ }\mu\text{sec}$ are used in an experiment. The large plate currents ($\sim 3\text{ }A_{\text{rms}}$ /tube) resulting when the grid bias is gated to $\sim 0\text{ V}$ can be sustained by the tube cathode for only short periods of time without significant degradation of the cathode.

A possible solution to this problem is to use a tube with a larger and more durable cathode such as the 7213 or 7214. This possibility is currently being pursued.

Another solution, which we use in our unit, is to gate the tube grids to only ~ -30 V for long rf pulses. While this solution reduces the rf power output (to ~ 1.2 kW), the plate current is reduced to less than ~ 1 A/tube. The change in grid bias during the rf pulse may be accomplished by tying the diode string in the collector circuit of Q_4 (Fig. 8) to -30 V rather than ground.

VI. PROBE

The spectrometer has been tested and found to work well with both resonant and delay line sample coils. The probe design, as well as the manner of using diodes and

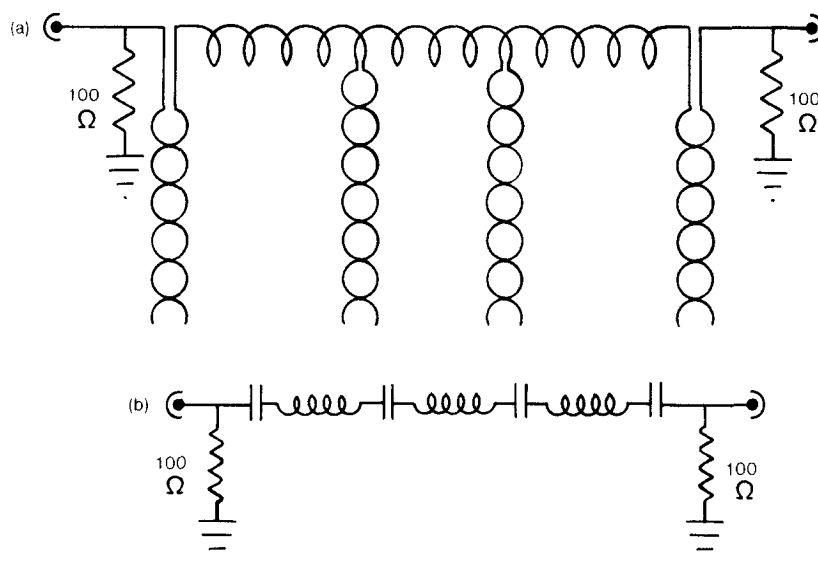


FIG. 10. (a) Construction of "pigtail" probe, (b) electrical equivalent of (a).

quarter-wavelength coaxial sections for connecting the gated power amplifier, probe, and receiver have been described in detail elsewhere (12–14).

The following section, which discusses the receiver, focuses attention primarily on recovery time following an rf pulse. The probe used for the recovery time measurements of the next section is sketched in Fig. 10.

The probe was constructed by winding coils of 28 AWG Teflon-coated wire around an ~ 5 -mm Teflon form. Three short (15 turns) coils were connected by twisting the wire at the ends of the coil sections together to form "pigtails" (Fig. 10a), and the capacitance between these leads represents the capacitors between the coil sections of Fig. 10b. The probe was tuned to $\sim 50 \Omega$ at 60 MHz by varying the length (and, therefore, changing the capacitance) of the pigtail capacitors. The main advantage in using this type of probe, in addition to its ease of construction, is that the impedance of the probe remains between 50 and 100Ω regardless of frequency, and the probe remains well-matched to the power amplifier and receiver over a large bandwidth.

VII. SIGNAL AMPLIFICATION

One of the fundamental problems in pulsed NMR involves minimizing the recovery time of the receiver following the overload which occurs during the rf pulse. Various techniques, including active probe damping (15, 16), receiver blocking schemes (12, 17, 18), and fast recovery preamplifiers (19), have been reviewed in the literature. Although the arrangement used in this spectrometer is not very different than that used by Lowe and Tarr (12), it has been used to achieve much shorter ($\sim 0.5 \mu\text{sec}$) recovery times. The

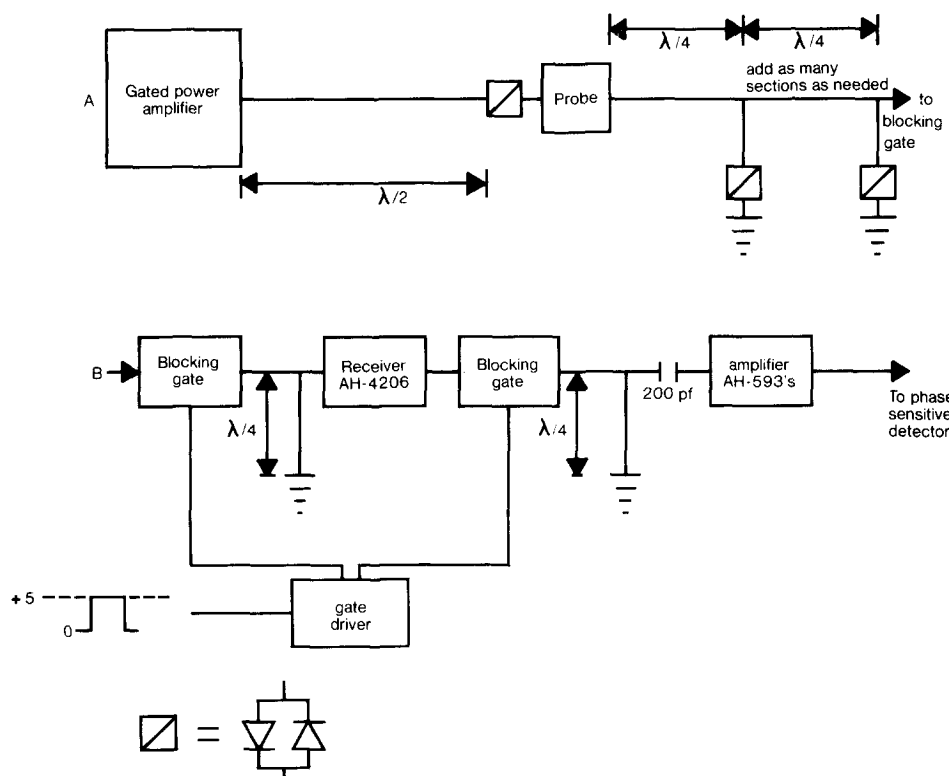


FIG. 11. (a) Coupling scheme of rf power amplifier, probe, and receiver; (b) block diagram of receiver and amplifier section.

receiver system is shown, in block form, in Fig. 11. Since the coupling of the probe to the gated power amplifier and receiver influences the recovery time, these components are also included.

The blocking gate and gate driver circuit assembly is shown in Figs. 12a and b, respectively. The gate, which was constructed by modifying a Hewlett-Packard Model 10534A double-balanced mixer using IN914B diodes as shown in Fig. 12a, achieves an isolation of ~ 60 db and an insertion loss of < 1.5 db. Modification of an HP mixer was chosen because the transformers contained by these mixers show almost no losses at 60 MHz, thus improving the overall insertion loss of the gate. It was found necessary to carefully match the diodes used in the construction of the gate in order to avoid gating pedestals and large switching transients.

The gate is driven by the circuit in Fig. 12b, which amplifies the receiver gating pulse from the PSE circuit of Section II. During an rf pulse, the circuit supplies ~ 60 mA to turn on diodes D_3 to D_6 of the gate in Fig. 12a, and diodes D_1 and D_2 are back biased by ~ 1.5 V. Between rf pulses, the circuit sinks current from diodes D_1 and D_2 which opens the gate and couples the probe and receiver. The small, variable RC network in the

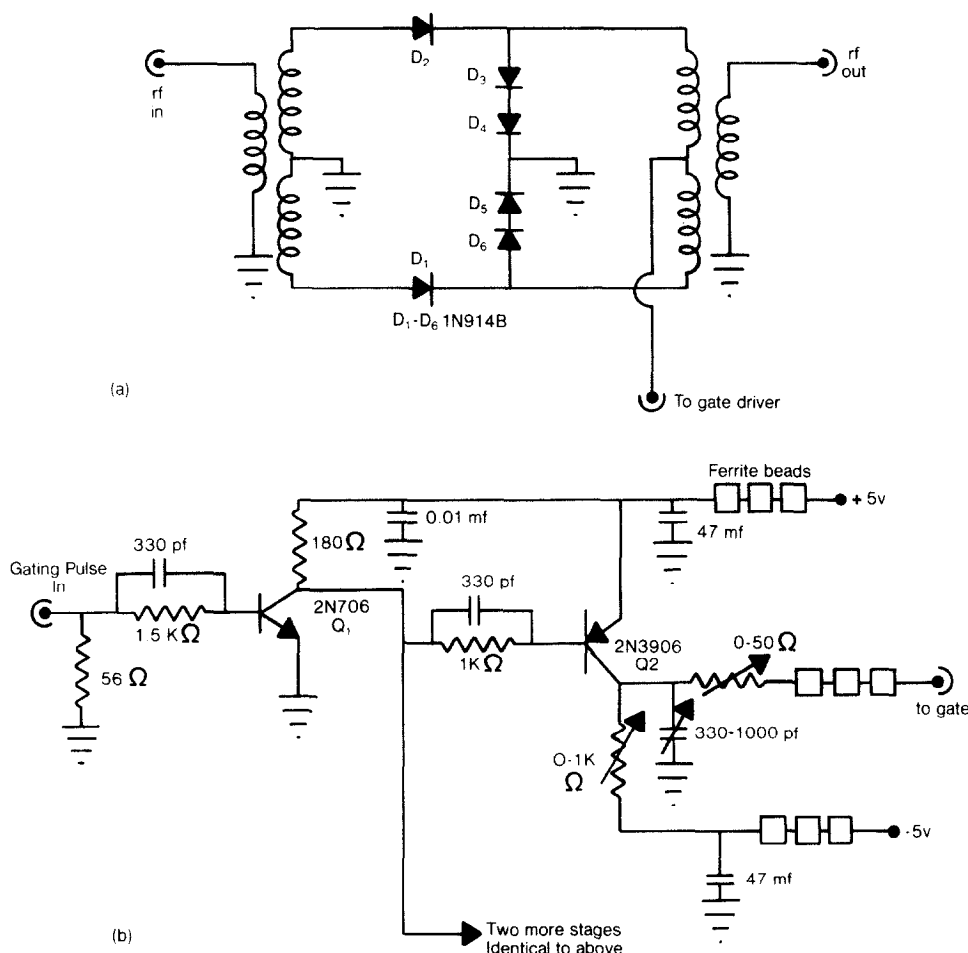


FIG. 12. (a) Diode blocking gate used before and after the receiver; (b) blocking gate driver circuit.

collector circuit of Q_2 (Fig. 12b) is used to slow the fall time of the pulse to the blocking gate and helps to reduce the amplitude of switching transients which may saturate the receiver following the gate. The gate driver has three outputs for driving other blocking gates between various stages of amplification, as shown in the block diagram (Fig. 11).

The receiver is a Model AH-4206 (Optimax, Inc.) amplifier with a bandwidth from 10 to 140 MHz, a gain of 32 db, and a noise figure of 2.3 db. An important feature of this amplifier is that it recovers quickly ($< 1.5 \mu\text{sec}$) from a 7-dbm overload. The receiver output is passed through another blocking gate, then further amplified by two (or more, depending on the gain desired) cascaded AH-593 amplifiers. The final signal from the

AH-593's is then detected by the phase-sensitive detectors (described in Section VIII). The quarter-wavelength shorts shown in Fig. 1 following each blocking gate aid in minimizing the switching transients and removes gating pedestals from the blocking gates. The small capacitor on the input of the AH-593 amplifier section was added because the AH-593's demonstrated poor transient response when the $\lambda/4$ short was connected directly to its input.

The overall recovery time of the signal amplifier system of Fig. 11 can be minimized by careful adjustment of the RC network in the output stage of the gate driver circuit. By setting the RC time-constant controlling the operation of the blocking gate following

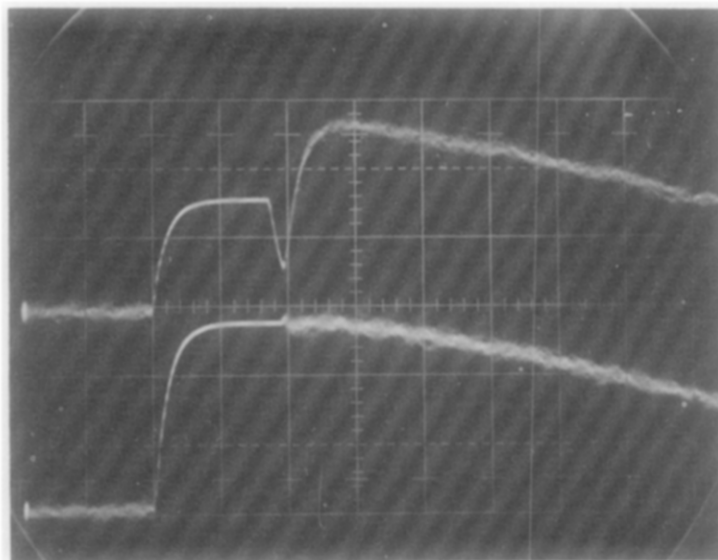


FIG. 13. Top: 60-MHz rf pulse from rf power amplifier; vertical scale, 500 V/cm; Bottom: output of phase sensitive detector showing initial portion of free induction decay (FID) from protons in H_2O doped with CuSO_4 ; vertical scale, 50 mV/cm. The horizontal scale for both traces is 1 $\mu\text{sec}/\text{cm}$. The wiggles on the proton FID is a ^{19}F signal from the Teflon coil form used in the probe.

the receiver to be slightly longer ($\sim 0.1 \mu\text{sec}$) than the one controlling the blocking gate before the receiver, the second gate also blocks the switching transients from the first gate after it has been amplified by the receiver. Both the overall recovery time of the signal amplification system and its bandwidth are illustrated in Fig. 13. The picture shows a detected signal from protons in CuSO_4 -doped H_2O , starting 0.5 μsec following the rf pulse. The wiggles superimposed on the H_2O free induction decay are the signal from the ^{19}F nuclei in the Teflon mandrel used as a coil form. The spectrometer for these measurements was set at 60 MHz for protons.

VIII. SIGNAL DETECTION

The spectrometer uses quadrature phase detection to obtain the final nuclear induction signal. The advantages of quadrature phase detection have been discussed elsewhere (7), and only a brief description of the circuitry used for this is presented here.

The complete phase detector assembly is shown in Fig. 14. The amplified nuclear induction signal (described in Section VII) passes through a blocking gate driven by a pulse from the PSE (Section II). This gate serves to block rf pulse leakage and amplified switching transients due to the receiver blocking gates of the preceding section from reaching the phase-sensitive detectors. The gated signal is split into two separate channels and then phase-detected using double-balanced mixers. The reference rf phases operating the detectors differ by 90° (described in Section IV), and the output of the phase detectors represent both the in-phase and out-of-phase components of the nuclear

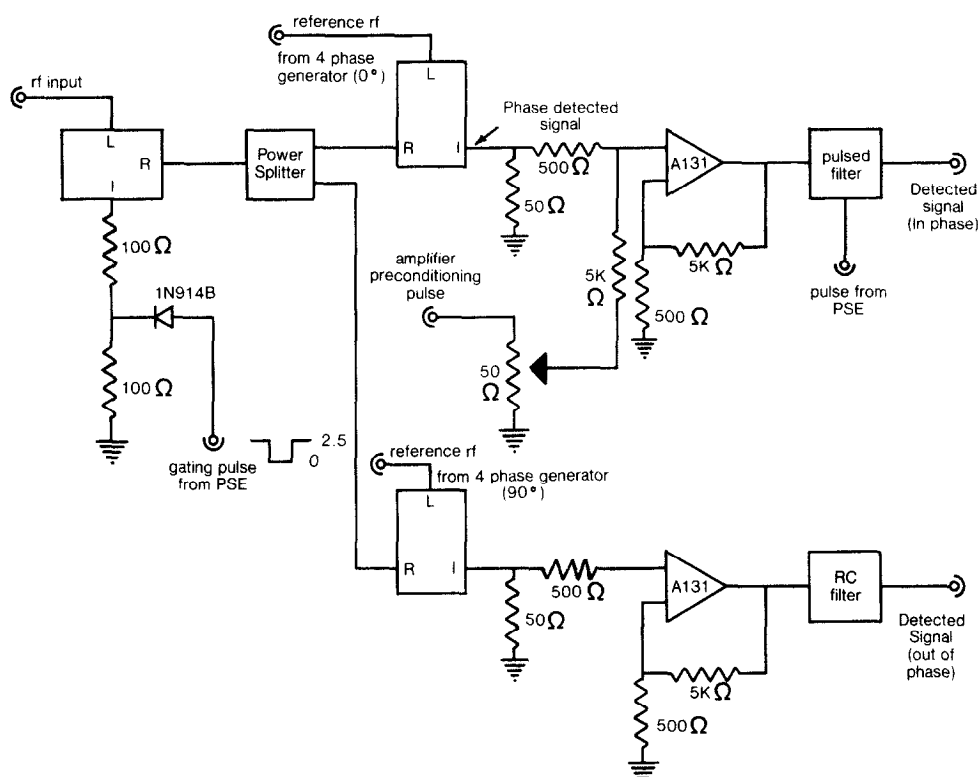


FIG. 14. Phase-sensitive detector assembly.

induction signal. The detected signals are amplified by operational amplifiers (Intech A-131; Intech, Inc., 1220 Coleman Ave., Santa Clara, Calif. 95050) and then filtered.

The portion of the phase detector assembly which handles the in-phase detected signal contains two features referred to in Fig. 14 as amplifier preconditioning and pulsed filter. These have proven useful in reducing distortion in the initial portion of the obtained free induction decay (fid) signal due to the limited bandwidth of the A-131 amplifier, filter, and various data-collecting devices (such as analog to digital (A/D) converters). Amplifier preconditioning involves the addition of a small pulse to the input of the A-131 amplifier just before the onset of the detected signal via the voltage divider network in Fig. 14. By setting the amplitude of the pulse to match the initial amplitude of the detected (fid) signal, the frequency response of the A-131 amplifier,

filter, and A/D converter distorts the rise time of the preconditioning pulse rather than the first several microseconds of the detected signal. The point at which the preconditioning pulse ends and the signal begins may be determined by adjusting the pulse amplitude to differ slightly from the initial signal amplitude.

The pulsed filter serves only to eliminate the distortion of the initial portion of the detected fid signal introduced by filtering and is not used when an amplifier preconditioning pulse is used. The pulsed filter, shown in Fig. 15, is operated by a pulse from the PSE and greatly shortens the time constant of the filter until after the onset of the detected fid signal. During the pulse, the fet (Q_3) in shunt with R of the RC filter conducts, representing a low impedance to the charging of the filter capacitor, C. After

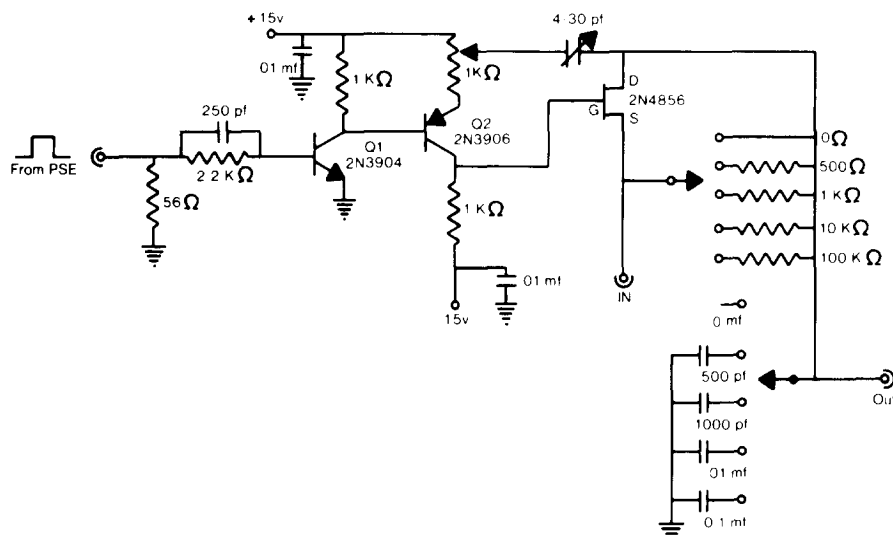


FIG. 15. Pulsed filter circuit.

the signal has obtained its maximum amplitude, Q_3 is turned off, thus lengthening the time constant of the filter. The pulsed filter thus permits filtering of the FID without severe distortion of its initial rise.

IX. TYPICAL OPERATION

This final section is concerned with some of the typical operating characteristics of the spectrometer described in Sections II to VII. Since the spectrometer does not contain any tuned circuit elements (with the exception of the probe), only three portions of the spectrometer require careful adjustment for optimum spectrometer operation. These devices are the PSE, the four-phase generator, and the receiver blocking gates; the adjustment of these devices is described below. The use of the amplifier preconditioning pulse and pulsed filter is also demonstrated. Finally, a brief discussion of the signal/noise (S/N) ratio to be expected from the spectrometer is presented.

As was mentioned in Section II, one of the purposes of the PSE is to compensate for the propagation delays in the rf gates, the power amplifier, and the receiver blocking gates. This compensation is to make certain that the start of the rf pulse from the rf

gates, and the third gating pulse of the rf power amplifier begin at the same time; and also that the receiver blocking gate is already in the blocking mode by the time the rf pulse begins. Appropriate compensation is accomplished by delaying the rf gating pulses to the rf gates until the grid gating pulse to the rf power amplifier is on, while not delaying pulses to the receiver blocking gate at all. The PSE generates the appropriate delays by using a certain number (as determined by the amount of delay desired) of TTL gates with a delay of ~ 10 nsec/gate, in the pulse channel to be delayed.

The four-phase generator (Section IV) generates rf of relative phases: 0, 90, 180, and 270° . It is recommended that the appropriate phase of each rf channel be verified, especially for the rf channels which drive the quadrature phase detector (Section VIII).

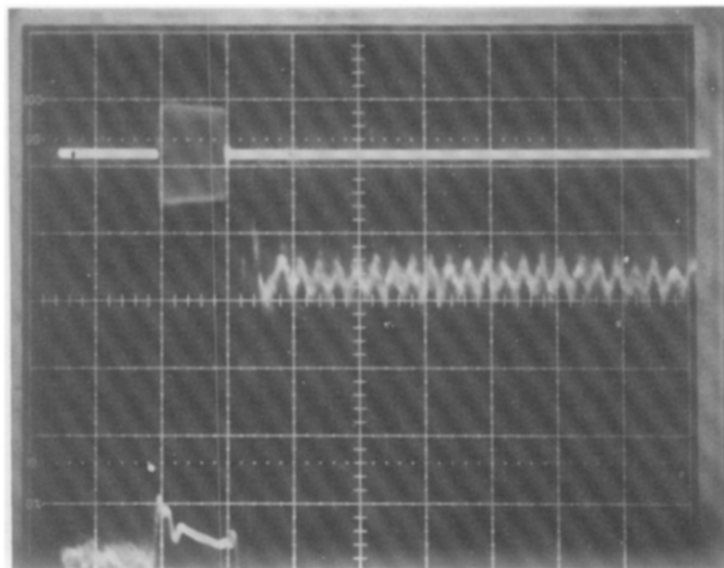


FIG. 16. Top: FID with amplifier preconditioning pulse preceding it, with the length and amplitude of the pulse not properly adjusted. Bottom: FID with properly adjusted amplifier preconditioning pulse. For both traces: vertical scale, 50 mV/cm, horizontal scale, 2 μ sec/cm. The FID is from protons in the metal hydride HfV_2H_3 . Note that an extra 1 μ sec is obtained at the beginning of the FID in the bottom trace.

Small corrections in phase of any channel may be made by adjusting the length of coaxial cable connecting the power splitters in the four-phase generator.

The receiver blocking network must also be adjusted for the shortest recovery time by adjusting the RC time constants in the output stages of the gate driver circuit (Section VII). This may be done by replacing the probe (see the block diagram, Fig. 1) by a $50\text{-}\Omega$ termination and observing the output of each amplifier section. The RC time constant of the driver circuit for the first blocking gate is set to be as small as possible, and the RC time-constant controlling the second blocking gate should be slightly longer, so that the amplifiers following the receiver are not exposed to the switching transients which result when the first blocking gate turns off. It should be mentioned that the blocking gates are "saturation prevention" rather than "receiver protection" devices, since the receiver may be operated continuously at the $\sim 7\text{-dbm}$ overload which would result if the $\lambda/4$ diode short (see Fig. 11a, Section VII) were connected directly to the receiver.

The use of the amplifier preconditioning pulse and pulsed filter are shown in Figs. 16 and 17, respectively. In Fig. 16, the preconditioning pulse is shown before amplitude (Section VIII) and length (Section II) adjustment (top trace), and after the proper adjustment (bottom trace). Figure 17 shows the use of the pulsed filter on a signal to which noise was added. Shown (from top to bottom) are the noisy FID without filtering,

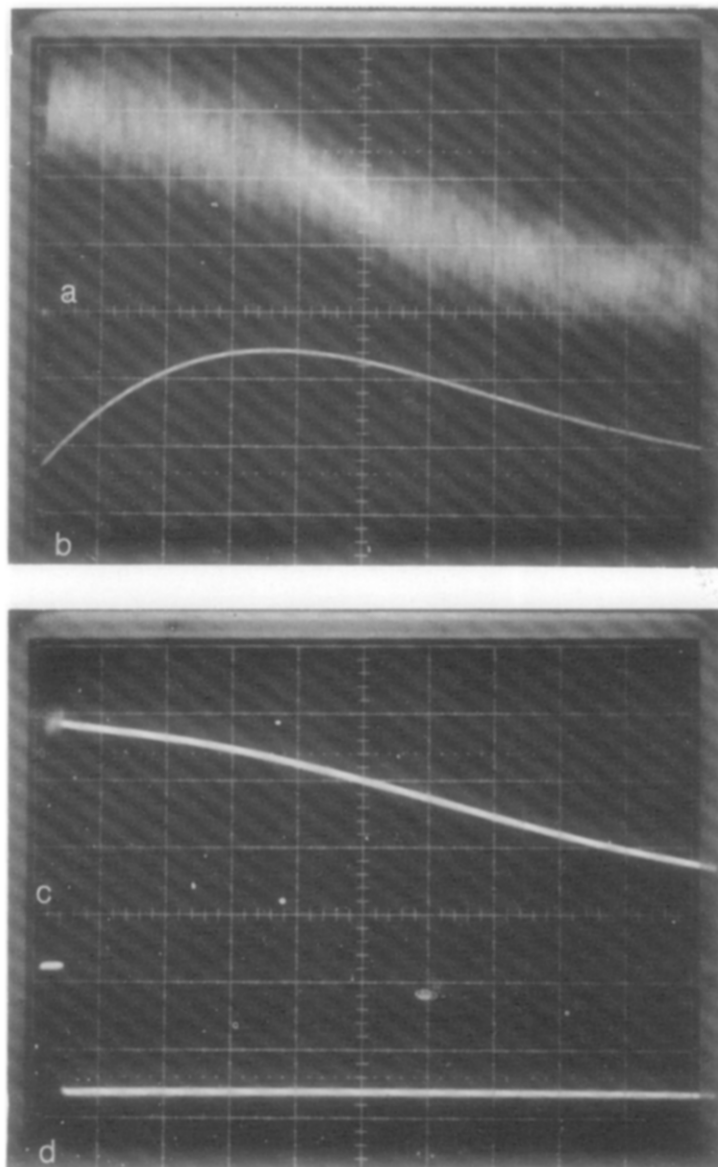


FIG. 17. a: noisy FID from protons in FeCl_3 -doped ethanol. b: same signal as above, heavily filtered; $\tau_{\text{RC}} \sim 500 \mu\text{sec}$. c: filtered signal obtained from pulsed filter. d: control pulse for pulsed filter from PSE. The vertical scale is 50 mV/cm, for the top three traces, and 1 V/cm for the bottom trace. For all four traces, horizontal scale: 2 $\mu\text{sec/cm}$.

the FID with filtering, the FID with pulsed filtering, and the pulsed filter controlling pulse. The amplifier preconditioning pulse and the pulsed filter essentially serve the same purpose: They both eliminate distortion of the rise of the FID caused by the low upper bandwidth of the devices (audio amplifier and filter) following phase detection (Section VIII). The amplifier preconditioning pulse technique is more useful because it can also eliminate similar distortion in the rise of the FID due to the low upper bandwidth of the A/D converters used in this laboratory for recording FID lineshapes.

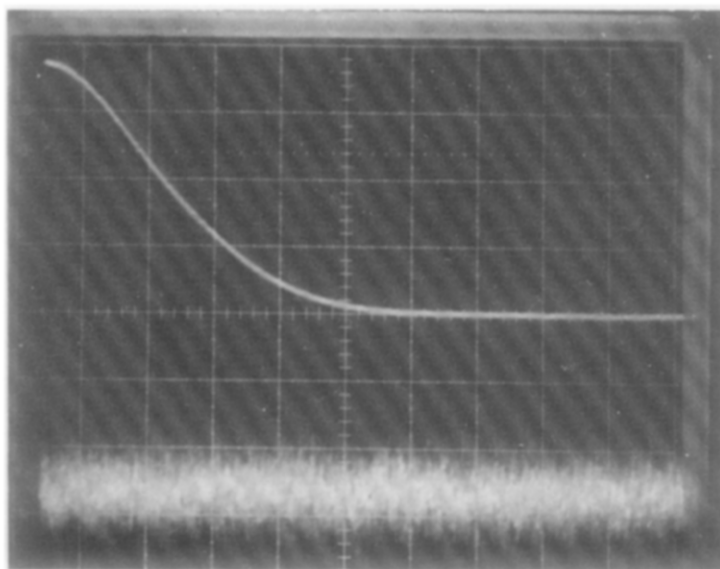


FIG. 18. Top: FID from protons in ethanol doped with FeCl_3 ; vertical scale, 100 mV/cm. Bottom: noise present on FID of top trace; vertical scale, 5 mV/cm. For both traces, horizontal scale: 500 $\mu\text{sec}/\text{cm}$. Both traces are single repetitions at full bandwidth ($\sim 2\text{MHz}$).

The topic of the (S/N) ratio of the spectrometer described above shall be treated only briefly and consists of a crude calculation of the expected (S/N) ratio. The (S/N) ratio actually obtained may be judged, more or less, from Fig. 18. The theoretical formula for the expected (S/N) ratio is given by the relation (20):

$$\left(\frac{S}{N}\right) = \eta M_0 \left[\frac{M_0 Q \omega_0 V_c}{4F k T_c \Delta f} \right]^{1/2}$$

η = fraction of sample coil volume occupied by sample

M_0 = magnetization of the sample in J/Tm³

$$= \frac{N(I)(I+1)\gamma^2 \hbar^2 B_0}{3kT}$$

N = number of spins per cubic meter

I = spin number of nucleus

γ = gyromagnetic ratio

B_0 = magnetic field

- k = Boltzmann's constant
 T = temperature of sample
 Q = the "Q" of the probe circuit
 $\omega_0 = 2\pi \cdot$ (the spectrometer operating frequency)
 V_c = volume of the sample coil
 F = ratio of (noise power out/noise power in) of receiver = $10^{f/10}$
 f = noise figure of receiver in decibels
 T_c = temperature of the sample coil
 Δf = the bandwidth of the detection system

The approximate values of the variables listed above, for the experimental conditions under which Fig. 18 was obtained, are listed below.

η	M_0	Q	ω_0	V_c	F	T_c	Δf
0.4	4×10^{-3} J/Tm ³	5	60 MHz	2×10^{-7} m ³	1.8	300°K	2 MHz

The value of M_0 listed above was calculated for $B_0 = 1.4$ T and the proton density in ethanol. The Q of the probe circuit was calculated from the shape of the rf pulse at the probe, which is given by

$$V = V_0 \exp -(\omega_0 t / 4\pi Q),$$

where V_0 = peak voltage of rf pulse. F was computed from the noise figure specified for the Optimax AH-4206 (~2.5 db), and Δf is the bandwidth of the A-131 operational amplifiers in the phase detector assembly. Using the values listed above, the expected (S/N) is ~150.

The (S/N) obtained in Fig. 18 can be estimated from the ratio of peak amplitude of the FID in the top trace to the estimated rms noise voltage in the bottom trace, and is ~100. This comparison of obtained and theoretical (S/N) is rather crude but demonstrates a reasonable agreement between the spectrometer's actual and theoretical (S/N).

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